SLoTAS: A Smooth Learnable Activation Function and a Controlled Re-evaluation of LoTAS

***Abstract*—Activation functions influence signal propagation, gradient flow, and the optimization of deep neural networks. This work examines the formulation and experimental evaluation of LoTAS, an activation function combining a logarithmic–tanh term, a sigmoid gate, and a linear residual. We introduce SLoTAS, a modified learnable activation that uses a convex residual mixture, a smooth input-magnitude term, and trainable scalar parameters constrained through softplus and sigmoid reparameterizations. We evaluate the proposed function on CIFAR-10 using a SimpleCNN under two training protocols: a paper-style protocol and a controlled protocol in which all evaluated activations use the same batch-normalization placement. Preliminary experiments use three random seeds. Under the controlled protocol, SLoTAS obtains a mean test accuracy of $89.98\%\pm0.11\%$, compared with $89.18\%\pm0.18\%$ for the evaluated LoTAS implementation. However, GELU achieves the highest mean accuracy of $90.41\%\pm0.13\%$ among the evaluated activation functions. The results indicate that SLoTAS is competitive under the current experimental setting but do not establish general performance superiority. The study is limited to one dataset, one architecture, three random seeds, and a 30-epoch training schedule.**

***Index Terms*—activation function, learnable activation, gradient flow, reproducibility, ablation study**

I. Introduction

The choice of activation function affects the nonlinear transformations learned by a neural network and can influence optimization, gradient propagation, and generalization. ReLU is widely used because of its computational simplicity, while alternatives such as Leaky ReLU, PReLU, ELU, GELU, SiLU, Mish, and SERF have been proposed to address different limitations of conventional rectifiers.

Recently, LoTAS was introduced as a composite activation involving a logarithmic–tanh term, a sigmoid gate, and a linear residual. Its reported performance motivates a careful examination of its mathematical formulation and experimental evaluation.

This work introduces SLoTAS, a modified learnable activation inspired by the structure of LoTAS. The proposed formulation uses a convex mixture between a residual component and a gated nonlinear component. It also replaces the absolute-value input term with a smooth positive surrogate and makes three scalar parameters trainable.

The proposed function is evaluated on CIFAR-10 using a common architecture and multiple random seeds. In addition to comparing SLoTAS with conventional activation functions, we evaluate it against an implemented LoTAS formulation under a controlled training protocol. The experiments are intended as a preliminary evaluation rather than a claim of general superiority.

The main contributions of this work are summarized as follows:

- A modified learnable activation function, SLoTAS, is introduced using a convex residual mixture and a smooth input-magnitude transformation.
- The mathematical formulation, multiplier bounds, derivative, and smoothness properties of the proposed function are analyzed.
- A controlled CIFAR-10 evaluation is conducted using a common SimpleCNN architecture and consistent batch-normalization placement.
- An ablation study investigates the contribution of the gating mechanism and other formulation components.
- The experimental results are analyzed with attention to protocol dependence and the limitations of the current evaluation.

II. Related Work

Activation functions include fixed rectifiers and their variants, such as ReLU, Leaky ReLU, PReLU, and ELU, as well as smooth nonlinearities such as GELU, SiLU, Mish, and SERF.

Learnable activation functions introduce trainable parameters that allow the activation response to adapt during training. These functions provide additional flexibility compared with fixed activation functions, although their effectiveness depends on the model architecture, parameterization, and optimization procedure.

The proposed SLoTAS belongs to the family of parameterized composite activations. Its formulation is related to LoTAS in its use of a logarithmic–tanh transformation and sigmoid gating, but differs in the way the residual term is combined with the gated component and in the parameterization of the input-magnitude term.

III. Analysis of the Base LoTAS Formulation

The base LoTAS formulation considered in this work is

$$f_{\mathrm{LoTAS}}(x) = x \tanh\left(\ln(1+|x|^{\alpha})\right)\sigma(\beta x) + \gamma x, \qquad (1)$$

where $\sigma(z) = 1/(1+e^{-z})$ is the sigmoid function.

For positive inputs with sufficiently large magnitude, the logarithmic–tanh factor approaches one and the sigmoid gate approaches one. Consequently, the positive tail behaves approximately as $(1+\gamma)x$. For large negative inputs, the sigmoid gate approaches zero, and the negative tail behaves approximately as γx.

The differentiability of the term $|x|^\alpha$ at the origin depends on the value of α. Therefore, smoothness claims about the base formulation must specify the parameter setting being considered.

In addition, comparisons between activation functions can be affected by differences in normalization, optimization settings, training duration, and model architecture. These factors motivate the controlled protocol described in Section V.

IV. Proposed Activation Function: SLoTAS

A. Definition

We define SLoTAS as

$$f_{\mathrm{SLoTAS}}(x) = x\left[\gamma + (1-\gamma)h(u)\sigma(\beta x)\right], \tag{2}$$

where

$$u = (x^2+\varepsilon)^{\alpha/2}, \tag{3}$$

and

$$h(u) = \tanh\left(\ln(1+u)\right) = \frac{u(u+2)}{u^2+2u+2}. \tag{4}$$

The parameters are constrained by

$$\alpha = \mathrm{softplus}(a), \qquad \beta = \mathrm{softplus}(b), \qquad \gamma = \sigma(c), \tag{5}$$

where a, b, and c are trainable scalar parameters for each activation layer. The smoothing constant is set to $\varepsilon = 10^{-4}$ in the reported experiments.

The initial parameter values are

$$(\alpha, \beta, \gamma) = (1.0, 1.5, 0.25). \tag{6}$$

B. Analytical Properties

To analyze the behavior of the proposed SLoTAS activation, we define the multiplicative term as

$$m(x) = \gamma + (1-\gamma)h(u)\sigma(\beta x), \tag{7}$$

where

$$u = (x^2+\varepsilon)^{\alpha/2}, \tag{8}$$

and $\sigma(z) = 1/(1+e^{-z})$ denotes the sigmoid function. Thus, the proposed activation can be expressed as

$$f_{\mathrm{SLoTAS}}(x) = xm(x). \tag{9}$$

Since $\varepsilon > 0$, we have $u > 0$ for every $x \in \mathbb{R}$. Furthermore, for $u > 0$, the function $h(u)$ satisfies

$$0 < h(u) < 1, \tag{10}$$

while the sigmoid function satisfies

$$0 < \sigma(\beta x) < 1. \tag{11}$$

Given the parameter constraint $0 < \gamma < 1$, the multiplier is therefore bounded as

$$\gamma < m(x) < 1, \qquad x \in \mathbb{R}. \tag{12}$$

This establishes that the multiplicative factor remains within a positive bounded interval.

1) First Derivative: Using the product rule on $f_{\mathrm{SLoTAS}}(x) = xm(x)$, the first derivative is

$$f'_{\mathrm{SLoTAS}}(x) = m(x) + xm'(x). \tag{13}$$

Differentiating the multiplier in Eq. (7) with respect to x gives

$$m'(x) = (1-\gamma)\Big[h'(u)u'(x)\sigma(\beta x) + h(u)\beta\sigma(\beta x)\big(1-\sigma(\beta x)\big)\Big]. \tag{14}$$

The derivative of u with respect to x is

$$u'(x) = \alpha x(x^2+\varepsilon)^{\frac{\alpha}{2}-1}. \tag{15}$$

Similarly, differentiating

$$h(u) = \frac{u(u+2)}{u^2+2u+2} \tag{16}$$

with respect to u yields

$$h'(u) = \frac{4(u+1)}{(u^2+2u+2)^2}. \tag{17}$$

Substituting Eqs. (15) and (17) into Eq. (14), and then into Eq. (13), provides the analytical expression for the first derivative of SLoTAS.

2) Smoothness and Derivative Behavior: For fixed $\varepsilon > 0$, the term $(x^2+\varepsilon)^{\alpha/2}$ is smooth for all $x \in \mathbb{R}$ when $\alpha > 0$. The functions $h(u)$ and $\sigma(\beta x)$ are also smooth over their respective domains. Consequently, the proposed activation is smooth for finite positive α and β, with $0 < \gamma < 1$.

At the origin, the activation satisfies

$$f_{\mathrm{SLoTAS}}(0) = 0. \tag{18}$$

Moreover, its partial derivatives with respect to the trainable parameters a, b, and c are finite and equal to zero at $x = 0$, owing to the multiplicative factor x in the activation definition.

Although Eq. (12) establishes that the multiplier is bounded, it does not, by itself, establish a global bound on the first derivative because the derivative also contains the term $xm'(x)$. Therefore, the boundedness of the multiplier should not be interpreted as a proof of globally bounded slope.

V. Experimental Setup

Dataset: CIFAR-10 is used for the preliminary evaluation. The training data are divided into a stratified 90% training subset and a 10% validation subset. Input normalization uses statistics calculated from the training subset. The test set is used for evaluation.

Architecture: A SimpleCNN is used, consisting of five 3×3 convolutional layers arranged in three stages and one hidden dense layer. The architecture is held constant across activation-function comparisons. Since the layer configuration of the original LoTAS experiments is not fully specified in the available implementation details, exact numerical reproduction is not claimed.

Activation functions: The evaluated functions include ReLU, Leaky ReLU, PReLU, ELU, GELU, SiLU, Mish, SERF, LoTAS, and SLoTAS. SiLU is listed once; it is equivalent to Swish with $\beta = 1$.

Training protocols: Two protocols are considered.

- `paper`: SGD with initial learning rate 0.01, momentum 0.9, weight decay 5×10^{-4}, batch size 64, no data augmentation, and a step schedule that halves the learning rate every two epochs. In this protocol, the LoTAS implementation follows the normalization arrangement used in the base-paper implementation, while the baseline activations are evaluated without that additional normalization arrangement.
- `fair_bn`: all activations are evaluated in a Conv–BatchNorm–Activation arrangement. SGD with Nesterov momentum, cosine learning-rate decay, flip/crop augmentation, and batch size 128 is used.

Both protocols use 30 training epochs in the current experiments. This is shorter than the training schedule reported in the base LoTAS study and should be considered when interpreting the results.

Metrics: The saved run outputs include test accuracy, macro-F1, macro one-vs-rest AUC, test loss, expected calibration error (ECE), per-class recall, gradient statistics, training time, and learned activation parameters. The main comparison reports accuracy, macro-F1, and AUC. Additional metrics are discussed only when their summaries have been verified.

Reproducibility: The reported fair-BN classification summaries are calculated over three random seeds. The implementation uses TensorFlow/Keras. The notebook, configuration files, and raw run logs should be included as supplementary material after the final experiment audit.

VI. Results and Discussion

This section presents the preliminary evaluation of SLoTAS on CIFAR-10 using the SimpleCNN architecture. The reported fair-BN results are based on three random seeds. Since the experiments use a 30-epoch training schedule, they should be interpreted as preliminary findings under the specified protocols, not as an exact reproduction of the original LoTAS study.

A. Comparison under the Controlled Fair-BN Protocol

The controlled comparison places each activation after batch normalization in the same Conv–BatchNorm–Activation arrangement. Table I reports the mean test accuracy, macro-F1, and macro one-vs-rest AUC with sample standard deviations over three seeds.

Under this configuration, SLoTAS achieves a mean test accuracy of $89.98\% \pm 0.11\%$, compared with $89.18\% \pm 0.18\%$ for the evaluated LoTAS implementation. This corresponds to an absolute improvement of 0.80 percentage points in the current experimental setting.

The proposed SLoTAS also achieves higher mean accuracy than ReLU, Leaky ReLU, PReLU, and ELU. However, GELU, SiLU, Mish, and SERF achieve higher mean accuracy values. GELU obtains the highest mean accuracy of 90.41% and macro-F1 of 90.39% among the evaluated functions.

The difference between SLoTAS and GELU is approximately -0.43 percentage points in mean accuracy. These results indicate that the proposed modification is competitive with the evaluated LoTAS implementation under the controlled protocol, but they do not establish general superiority over LoTAS or other activation functions.

Additional seeds, architectures, datasets, and longer training schedules are required to assess the robustness and generality of these observations.

B. Ablation Study

An ablation study investigates the contribution of individual components of SLoTAS. The evaluated configurations include the complete trainable formulation, a fixed-parameter configuration, a configuration without the gating mechanism, a configuration without the logarithmic–tanh component, and a configuration without the leakage contribution.

The complete trainable SLoTAS configuration achieves 89.98% mean accuracy, compared with 89.26% for the fixed-parameter configuration. Removing the gating mechanism reduces mean accuracy to 87.77%, a decrease of approximately 2.21 percentage points relative to the complete formulation.

In contrast, removing the logarithmic–tanh component results in a mean accuracy of 90.05%, while removing the leakage contribution results in 90.02%. These values are marginally higher than the complete formulation. The differences are small and should not be interpreted as evidence that the removed components are unnecessary. More seeds and controlled statistical analysis are needed.

The hard-absolute-value configuration produced invalid training behavior in all three runs. These runs are treated as failed configurations rather than valid classification results. The numerical cause of this failure has not yet been established and requires further investigation.

The ablation results suggest that the gating mechanism may contribute to the observed performance in this particular setup. The individual effects of the other components remain inconclusive.

TABLE I
CIFAR-10 RESULTS USING SIMPLECNN UNDER THE FAIR-BN PROTOCOL. VALUES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER THREE RANDOM SEEDS.

Activation	Accuracy (%)	Macro-F1 (%)	Macro-AUC (%)
ReLU	89.88 ± 0.22	89.86 ± 0.22	99.35 ± 0.03
Leaky ReLU	89.95 ± 0.05	89.93 ± 0.05	99.36 ± 0.03
PReLU	89.95 ± 0.24	89.94 ± 0.24	99.37 ± 0.03
ELU	88.68 ± 0.15	88.65 ± 0.16	99.23 ± 0.03
GELU	$\mathbf{90.41 \pm 0.13}$	$\mathbf{90.39 \pm 0.12}$	$\mathbf{99.42 \pm 0.02}$
SiLU	90.27 ± 0.21	90.25 ± 0.20	99.41 ± 0.03
Mish	90.20 ± 0.04	90.18 ± 0.05	99.40 ± 0.02
SERF	90.03 ± 0.13	90.02 ± 0.12	99.39 ± 0.03
LoTAS	89.18 ± 0.18	89.16 ± 0.19	99.28 ± 0.01
SLoTAS	89.98 ± 0.11	89.98 ± 0.11	99.38 ± 0.01

TABLE II
PRELIMINARY ABLATION RESULTS ON CIFAR-10 USING SIMPLECNN UNDER THE FAIR-BN PROTOCOL. VALUES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER THREE SEEDS.

Configuration	Accuracy (%)	Macro-F1 (%)
Full SLoTAS	89.98 ± 0.11	89.98 ± 0.11
Fixed parameters	89.26 ± 0.10	89.23 ± 0.11
No gate	87.77 ± 0.31	87.74 ± 0.32
No logarithmic–tanh	90.05 ± 0.29	90.04 ± 0.29
No leakage contribution	90.02 ± 0.09	90.01 ± 0.09

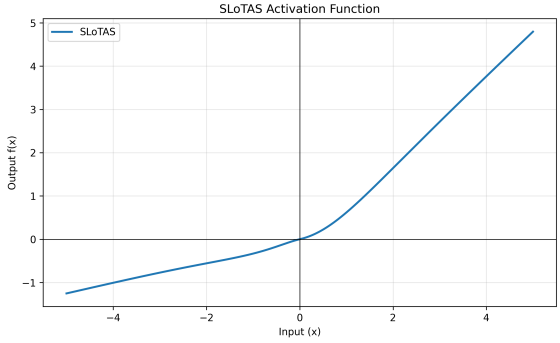


Fig. 1. SLoTAS activation curve using the initial parameter configuration.

C. Activation and Derivative Analysis

The activation output and first derivative were evaluated numerically over a finite input interval using the initial parameter configuration $\alpha = 1.0$, $\beta = 1.5$, $\gamma = 0.25$, and $\varepsilon = 10^{-4}$.

Within the sampled interval $[-5, 5]$, the numerical derivative had a minimum value of approximately 0.2055 and a maximum value of approximately 1.0757. These values describe only the evaluated interval and parameter configuration; they do not establish a global derivative bound.

The plots provide a preliminary visualization of the activation response. Further analysis should examine the learned parameter values after training and verify the derivative over a wider input range and across multiple parameter settings.

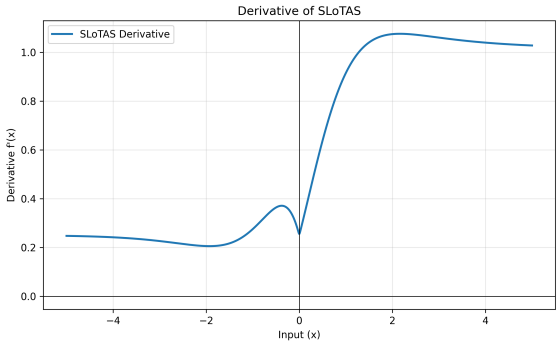


Fig. 2. Numerically evaluated first derivative of SLoTAS using the initial parameter configuration.

D. Training Stability and Protocol Dependence

The preliminary experiments indicate that the observed performance of SLoTAS depends on the training protocol. In the paper-style protocol, the initial SLoTAS run failed to learn meaningful class distinctions and remained near chance-level accuracy. In contrast, the fair-BN experiments achieved substantially higher classification accuracy.

This difference indicates that the interaction among the activation formulation, normalization strategy, and optimization configuration requires further investigation. It also demonstrates why activation comparisons should control normalization and other training conditions.

The paper-style protocol should not be interpreted as a direct performance comparison between all activation functions because the normalization arrangements differ. The fair-BN protocol provides the more controlled comparison for the current study.

E. Limitations

VII. CONCLUSION

REFERENCES